

Exact Audits of Worked Systems for the Obstruction Calculus: Counts, Witnesses, and Monitoring Design Rules

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Abstract

Viability certificates are useful in application to the extent that their behaviour on concrete systems is known exactly. This paper develops one exact audit frame — integer and rational arithmetic throughout, every count regenerated by script — and applies it to a family of worked systems that instantiate the mechanisms of the obstruction calculus (Abaee, 2026b): a shared finite two-coordinate system with observation structures (aggregation, policy-class restrictions, decentralized observation, regime uncertainty), a review-timing grid for a hidden-regime resource model, and a continuous two-floor production benchmark. The audits yield the following. The review-timing identity on the 48-cell grid is non-strict: viability at the deadline holds if and only if $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, boundary included; the strict form fails at exactly one cell. On the four-state monitoring target, 7 of the 15 partitions are adequate with two minimal two-cell partitions, and the two maximal common-action sets overlap, so no coarsest adequate design exists. Four policy classes have kernel sizes 24, 26, 25, and 28 of 36 belief pairs, forming a Boolean product with incomparable middle cells. Decentralized unanimity over a 256-law space sustains 12 of 36 pairs; an authority restriction collapses the kernel to zero. Mode-blind operation loses exactly the pairs containing one state. The biased certainty-equivalence law has drift at least 21/100 on $[1, 2]$; the bias-free law has drift identically zero. The benchmark’s critical aggregate is $Y^* = 27/5$, with a rational dual certificate of margin 3/50. Static duality holds at value $-1/10$ on the two-floor instance with a normalized Farkas pair, and a Helly-tight triple exhibits the sparse $m+1$ -point witness. Design rules follow: review deadlines are boundary-inclusive; merged fibres call for splits, not threshold tuning; aggregated indices support certainly-safe reporting only; biased observation maps call for structural correction.

Keywords: viability kernels; obstruction certificates; exact arithmetic; monitoring design; decentralized observation; certainty equivalence

1. Introduction

The obstruction calculus states sound sufficient conditions for nonviability — obstruction certificates — mech-

anism by mechanism: common-action failure, review-timing failure, tube failure, finite-time exit, certification failure through fibre crossing, and failure induced by restricting the policy class (Abaee, 2026b). Each mechanism is accompanied in application by quantitative questions. How many belief states does an observation structure destroy, and which ones? At which review deadlines does the timing certificate fire, and does it fire at the boundary? What exactly does a vote protocol cost, and which law in the protocol’s space sustains a given target? This paper answers such questions exactly for a family of worked systems, under one audit frame and one notation.

Three properties define the frame. First, all computations are carried out in integer and rational arithmetic: no solver, no tolerance, no simulation. Second, every system in the family is finite and every count is regenerated by a deterministic standard-library script, so any two executions agree. Third, each audited quantity is stated together with a witness or a certificate — an action sequence, a partition, a law pair, a dual measure — so the counts carry their own evidence.

The systems are drawn from the applied model of Abaee (2026a), in which a two-coordinate resource stock is managed by acting one of two instruments under incomplete observation. The family comprises: the shared audit system with its observation structures (Sections –); the hidden-regime review grid (Section); a continuous two-floor production benchmark (Section); and a multiple-floor line extending the Farkas certificate to jointly binding constraints (Section).

Systems, conventions, and methods

The shared audit system. States are pairs $x = (z_1, z_2) \in \{0, 1, 2, 3\}^2$; the safe set is $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$ with the two coordinates as floors. Instruments are $u \in \{0, 1, 2\}$: $u = j \in \{1, 2\}$ acts coordinate j , which regenerates one unit up to the cap 3, while the other coordinate loses one unit; $u = 0$ executes no instrument and both coordinates lose one. The corner $(1, 1)$ is the single state from which no instrument maintains both floors, so its singleton is nonviable; the nine viable states generate $C(9, 2) = 36$ two-element belief pairs, the audited population, at review horizon $H = 12$ under the audited two-horizon verdict (the winning recursion at H and at $H - 2$; Abaee, 2026b,

Theorem 1).

Observation structures. An observation map $\text{info} : x \mapsto c$ partitions a belief into fibres; a policy acts per fibre on the read cell. The audited policy classes combine two aggregation levels (full information; aggregation $c(x) = z_1 + z_2$) with two protocols (free per-step choice; the no-repeat protocol that forbids repeating the previous action).

Exactness. Counts are computed by backward recursion over fibre partitions with exact state arithmetic; witnesses are extracted as step-one per-fibre action vectors along a winning branch. The census of Section enumerates all 15 partitions of a four-state target. The continuous objects of Section are rationals.

Conventions. On the hold class of the hidden-regime model, the discrete survival supremum is $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$, with level sets $\{z_0 \geq 1 + k\}$. Two certainty-equivalence laws are audited: an uncorrected law whose index leads the state by one step, $s \mapsto s + 0.1$, and a corrected law with the lead removed; both are read through the index $g(s) = s^2$.

Review timing on the hidden-regime grid

The hidden-regime resource model is audited on the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, 48 cells. Viability at the deadline holds if and only if $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, with zero mismatches across the grid. The identity is boundary-inclusive: at the cell $(z_0, T_{\text{obs}}) = (2, 1)$, on the boundary $z_0 = 1 + T_{\text{obs}}$, the system is viable, and the strict form $T_{\text{obs}} < \sigma^*(z_0)$ fails at exactly this cell. The practical form of the rule: a review must land no later than the worst-case exit time, and a deadline exactly at the worst-case exit is admissible.

Monitoring adequacy

Let $R(x)$ denote the safe actions of state x : those keeping x in $\mathcal{V}$ for one step. A partition of a target belief is *adequate* when every fibre has a nonempty common safe-action set. On the four-state target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$, 7 of the 15 partitions are adequate, and two minimal adequate partitions with two cells exist.

The maximal subsets with a nonempty common safe-action set are $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$. They overlap in $(2, 2)$, so no partition into maximal common-action sets exists: coarsening to maximal fibres is not an available design, and the adequate partitions found by the census are the design options. Two consequences follow. Pairwise intersection of safe-action sets is necessary for fibre adequacy but not sufficient in general; and an observation map merging states with disjoint safe-action sets obstructs at that information state while every branch remains individually viable — the remedy is a fibre split, not a finer threshold on the same index.

Policy classes

The four audited classes combine full information and aggregation with free and no-repeat protocols. Their kernels among the 36 pairs have sizes:

$$\begin{aligned} |\mathcal{W}_{\text{inst}}| &= 24, & |\mathcal{W}_{\text{agg}}| &= 26, \\ |\mathcal{W}_{\text{full-codex}}| &= 25, & |\mathcal{W}_{\text{full}}| &= 28. \end{aligned}$$

The four cells form a Boolean product with incomparable middle cells: the aggregate kernel contains pairs outside the full-information codex kernel (for instance $\{(1, 2), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$), and the codex kernel contains pairs outside the aggregate (for instance $\{(1, 3), (3, 1)\}$). Neither protocol dominance holds. The meet of the two middle kernels equals the institutional kernel setwise, and the join is the full-information kernel, which is exactly the $C(8, 2) = 28$ pairs avoiding the corner $(1, 1)$ — the corner’s singleton nonviability accounts for every pair-level loss at full information.

Decentralized observation

Two agencies each observe one coordinate and vote; an instrument executes only on agreement. Each agency’s vote is a map from its coordinate’s four values to the two instruments, so the law space has $2^4 \times 2^4 = 256$ law pairs. Unanimity sustains 12 of the 36 pairs, and exactly 16 coordinated-viable pairs are lost. The loss instance $\{(1, 2), (3, 1)\}$ is representative: each branch is individually sustainable, and no law pair sustains the belief. For the audited belief $\{(1, 2), (2, 1)\}$, exactly 16 of the 256 law pairs sustain it; the count coincides numerically with the 16 lost pairs, and no identity is implied. Forbidding agency 2’s vote (fixing it to instrument 1) collapses the kernel from 12 to 0: the second agency’s voice, not the first’s, carries the audited target. Two modelling choices are declared. The deadlock transition $u = 0$ executes no instrument and both coordinates lose one; and each agency votes on its own coordinate’s current value, so the protocol carries local memory with delay, not common knowledge.

Regime uncertainty

Two regimes alternate: an adversary chooses at the end of step $t - 1$ the regime realized at step t ; the acting regime damages its coordinate by two units, the other regenerates one. With frozen regimes the kernel is $36 = 36$: the corner $(1, 1)$ is viable under each regime separately. Mode-blind operation, with the regime read only through the state, yields a kernel of exactly 28 — every pair containing the corner is lost, no merged-pair loss exists, and the loss is generated entirely at the singleton level. The mode-blind kernel agrees setwise with the full-information kernel of Section ; the agreement is coincidental, arising under different damage routings, and no identity is claimed.

The continuous benchmark and static duality

Two stocks face demand floors under one control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ with caps $\text{cap}_1(Y) = 3/2 - (Y - 2)/10$ and $\text{cap}_2(Y) = 59/50 - (Y - 2)/10$. The critical aggregate is $Y^* = 27/5$: the cap sum equals 2 exactly there. The $Y = 5$ fibre is viable with witness $(6/5, 4/5)$; the $Y = 6$ fibre is obstructed with cap sum $47/25 < 2$, certified by the dual measure $(1/2, 1/2)$ on the two active-floor atoms with margin $3/50$. The fibre width is $\Delta x_1(Y) = Y - 4$.

Static duality on the two-floor instance: the max-min equals the min- λ max at value $-1/10$, with multiplier $\lambda = (1/2, 1/2)$ the normalized Farkas pair of the calculus’s worked certificate (Abaee, 2026b). Convexity is load-

bearing: with $U = \{0, 1\}$ and rows $(-1, +1)$, $(+1, -1)$, the common set is empty, the max-min is -1 , and the min-multiplier maximum is 0 . Under convexity the witness is sparse: the Helly-tight triple $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ attains the obstruction with $m+1 = 3$ points. Weak duality holds unconditionally; strong duality holds in the finite-horizon finite and static-observation classes.

Multiple floors

With several binding floors the Farkas certificate obstructs jointly: a single certificate over all floors certifies nonviability where per-floor certificates do not compose. On the audited two-floor extension, the minimal infeasible subsystem carries explicit witnesses, and the corner $(1, 1)$ is dynamically nonviable as in the shared system. Two further objects are recorded for subsequent study: a first-breach timing bound for multi-floor systems, and a corner-cone reading of the infeasibility region.

Design rules

Four rules hold as identities on the audited systems and transfer as default guidance. *Timing*: reviews are boundary-inclusive — $T_{\text{obs}} \leq \sigma^*(z_0)$, the deadline at worst-case exit admissible (Section). *Coarseness*: a fibre merging states with disjoint safe-action sets obstructs at that information state; the remedy is a fibre split, not a finer threshold on the same index (Section). *Aggregation*: an index whose fibres cross a floor cannot support an exact safe/unsafe verdict; the certainly-safe set is the honest report. *Bias*: the uncorrected law’s drift is at least $21/100$ on a 101-point grid of $[1, 2]$, while the corrected law’s drift is identically zero; a biased observation map empties a nonempty kernel under an injective index, and the remedy is structural correction of the map, not a better sensor. To these the decentralized audit adds a fifth: an authority restriction on one agent’s vote can collapse the sustainable set outright, so protocol design precedes instrument tuning (Section).

Verification methods

Every count, witness, identity, and bound in this paper is regenerated by a deterministic verification script in exact integer and rational arithmetic, standard library only. The script recomputes the counts of Sections – from the shared system’s primitives — the transition, the fibre partition function, and the backward recursion with the no-repeat protocol threaded as a parameter — enumerates the census of Section , evaluates the benchmark identities of Section in rational arithmetic, asserts the drift bounds, and checks that each headline number appears in the text. The audited identities are additionally covered by the per-system verification scripts of the companion archive (Abaee, 2026a; see the Code availability statement).

Conclusion

Exact audits bind the obstruction calculus to its worked systems: each mechanism acquires a count, a witness, and a boundary. The timing certificate fires everywhere

beyond $z_0 - 1$ and nowhere at the boundary; the adequate monitoring designs on the audited target are enumerable, and no coarsest one exists; policy-class value drops unequally across aggregation and protocol restrictions, with the two middle cells incomparable; unanimity costs a fixed and identified set of beliefs, recoverable only while both votes remain free; regime blindness loses exactly the corner’s pairs; and the continuous benchmark separates the viable from the obstructed at $Y^* = 27/5$ with rational certificates on both sides. The design rules — boundary-inclusive reviews, fibre splits over threshold tuning, certainly-safe reporting under aggregation, structural repair of bias, and protocol design before instrument tuning — are the audit’s operational content.

References

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- Abaee, A., 2026b. An obstruction calculus for viability under incomplete observation (submitted).

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_worked_systems_v3_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all counts, witnesses, and bounds verbatim.

AI declaration

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